

Four problems, and the scaffolding falls away

The symmetric ring of repressors, each gene shut off by the one behind it:

$$\frac{dx_i}{dt} = \frac{\alpha}{1 + x_{i-1}^n} - x_i, \quad i = 1 \dots N$$

with time in protein lifetimes and concentration in units of the repression threshold, so α and n are all that is left.

Today's three tools, and nothing else. At the symmetric fixed point $x_1 = \dots = x_N = x$:

$$\alpha = x(1 + x^n), \quad g = \frac{nx^n}{1 + x^n}, \quad J = -I - gP$$

P is the cyclic shift, so its eigenvalues are the N th roots of unity $\omega_k = e^{2\pi i k/N}$, and therefore

$$\lambda_k = -1 - g\omega_k, \quad \text{Re } \lambda_k = -1 - g \cos\left(\frac{2\pi k}{N}\right).$$

The state holds while every $\text{Re } \lambda_k < 0$; it fails when the worst one reaches zero. At each transition this sheet asks *why does that step follow?* Answer in writing before you go on.

1 — Fully worked: the ring of three at $n = 3$

The worst $\cos(2\pi k/3)$ is $-\frac{1}{2}$ (at $k = 1, 2$), so $\text{Re } \lambda = -1 + g/2$ and the state fails when $g = 2$.

Setting $g = 2$: $nx^n = 2(1 + x^n)$, so $x^n(n - 2) = 2$, giving $x^n = 2/(n - 2)$. At $n = 3$: $x^3 = 2$, so $x = 2^{1/3} = 1.260$, and

$$\alpha_c = x(1 + x^n) = 1.260 \times 3 = \mathbf{3.780}.$$

Below **3.780** the ring settles; above it, it oscillates. *Why does that step follow?* — because g rises with α , so crossing $g = 2$ once means crossing it for good.

2 — Last step blank: the ring of three at $n = 4$

Same three tools, one number changed. From $x^n = 2/(n - 2)$ with $n = 4$:

$$x^4 = \frac{2}{2} = 1, \quad \text{so} \quad x = \underline{\hspace{2cm}}$$

$$\alpha_c = x(1 + x^n) = \underline{\hspace{2cm}}$$

Why does *that step follow*? — is raising n from 3 to 4 making the oscillator easier or harder to build, and does that match what you would have guessed?

3 — Last two blank: a ring of five

Now $N = 5$ instead of 3. Nothing about the derivation changes except which roots of unity appear. The five values of $\cos(2\pi k/5)$ for $k = 0 \dots 4$ are

$$1, \quad 0.309, \quad -0.809, \quad -0.809, \quad 0.309.$$

The worst case is the most negative one, $-0.809 = -\cos(\pi/5)$, so

$$\text{Re } \lambda_{\text{worst}} = -1 + \underline{\hspace{1cm}} g \quad \implies \quad \text{the state fails when } g = \underline{\hspace{1cm}}$$

Why does that step follow? — a ring of five needs **less** gain than a ring of three. Say why in one sentence, in terms of what going once around the loop costs you.

And the one that is not symmetric. Try $N = 4$. What is the worst $\cos(2\pi k/4)$, and what is different about the eigenvalue that achieves it? What does that tell you about a ring with an even number of repressors?

4 — Bare problem

You have a working three-gene ring oscillator. Your collaborator needs it to **stop oscillating** and sit at a steady level — and they cannot change α , because the promoters are doing other work in the same cell.

Name a change you would make, say which quantity in today's derivation it moves, and say which way.

There is more than one right answer; give two if you can, and say which you would try first and why.
